Inference From a Miss: What Silence Licenses, and Two Corrections That Subtract

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Abstract

The owner of this project raised a specific worry about the engine's inference: if you ask for a low spade, miss, and the target never asks you for a low spade back, do not conclude they hold none. This paper answers that worry and then generalises it into the full matrix for choosing an ask. The engine does not make the inference and structurally cannot — an absence is not a log event, so nothing in the ingestion `switch` can key on one — demonstrated by a treatment/control pair in which only the named card's candidate set moves and the other four are bit-identical. The residual error runs *opposite* to the worry: in exactly the owner's position the engine believes 14.0% where the truth is 2.0% ($N = 19{,}003$), too generous to the silent seat. Calibration over 1,395,876 scored asks from 300 `us54` games puts the overall bias at -0.0109 — optimistic, which is the signature of an engine that is *not* over-inferring — with the certainty tier exact: 16,680 asks called certain hit 16,680 times. Stratified, silence alone is worth -0.0081 over 1,176,652 asks and is genuinely near-uninformative, while silence *after a miss* is a different object — such a seat holds a card of the set 6.3% of the time against 66.3% for a silent seat never missed on. Two real gaps run the other way, both under-reading *positive* evidence: a published row-6 licence is under-priced by 8.4 points (0.2386 believed against 0.3221 observed, $N = 23{,}345$), because a deal-time constraint is dropped the moment it is satisfied or exhausted — exactly when the seat has been shown to hold a card of the set; and the conceded-turn proxy is hand size, which correlates -0.147 with the cards the conceded seat then takes against $+0.421$ for its published reach. Conditioning on the licence at strength $\lambda = 0.60$ removes the calibration bias almost exactly (-0.0002). The headline is what that bought. On one 400-pair common baseline, defusal alone is worth $+1.915 \pm 0.315$ sets per duplicate pair, the calibration fix alone $+2.068 \pm 0.293$, and **both together** $+1.565 \pm 0.317$: the stack ranks below either part, which is the shape of two substitutes reached different ways rather than of complements. That ordering is *suggestive, not established*. Every interval quoted is an arm against the baseline; none is on a difference between arms, and the smaller gap the reading rests on (0.350 sets) is under these arms' own 0.42–0.47 detection floor. The arms do share one bank and one baseline, so the comparison is paired and sharper than a between-bank one would be — but the paired interval was not computed, and at this N the ordering is reported as a signal to re-measure rather than as a settled result. Only a common-baseline design raises the question at all — measured each against its own baseline the calibration fix reads as a flat $+0.04 \pm 0.41$. The same 400-pair scale is where this instrument is weakest, and the paper reports that in the body: the defusal quantity measures $+1.62/+1.66$ at 4,000 pairs, and the first explanation offered for the gap was tested and refuted. A separate observation — that two or three sets get "ask-traded" — is confirmed at 88.1% against a 21.4% availability-matched control and then shown not to be reciprocation at all: 99.1% of those replies are the hit-probability argmax. The recommendation is to ship neither change.

1 Introduction

Every ask in Literature (Canadian Fish) is a question put in public, and every answer is heard by the whole table. That makes the game a laboratory for a narrow question with a wide reach: *what does an event license you to conclude, and what does a non-event?*

The question arrived here in a concrete form. The project's owner, watching the engine play, wrote down a worry:

> "If you have a low spade, you ask for a low spade but don't get it, and they don't ask for a low spade back, don't automatically infer that they don't have any low spades."

It is a good worry. It names a real failure mode of enthusiastic deduction engines — the slide from "no evidence that they hold one" to "evidence that they hold none" — and it names it in the exact position where the slide is most tempting, because the miss *did* certify something and the silence sits right next to it.

This paper answers the worry, and then does the thing the worry was a special case of. It audits every path by which the engine eliminates a candidate and asks of each whether it is sound (Section 4). It measures whether the engine over-infers anywhere, by scoring its own stated hit probability against ground truth over 1,395,876 asks (Section 5). It measures what silence is actually worth, in the owner's exact position and in three neighbouring ones (Section 6). It reports the two places where the engine's inference *is* wrong — both of them the opposite of the feared error (Section 7) — and one behaviour that looks wrong from the outside and is not (Section 7.3). It lays out the full ask matrix the owner asked for, twelve factors with their grounding, sign, and whether they are priced today (Section 8). And it closes on the result that changed what the project would do (Section 9): fixing the calibration error is worth about two sets per duplicate pair, the shipped defusal heuristic is worth about two sets per duplicate pair, and *doing both measures lower than doing either* — an ordering the design makes visible and this bank size does not resolve.

Three findings are worth stating before the machinery.

The engine does not make the owner's mistake, and cannot. Not because it is careful, but because an absence is not representable in the layer that would have to act on it (Section 3). The measured residual runs the other way: the engine is too generous to the seat that has gone quiet, not too harsh.

Silence and silence-after-a-miss are different objects. The first is worth 0.8 points of bias over 1.18 million asks — the owner's principle, validated. The second is worth 60 points of set-level probability. The compound event is informative even though neither part is, and the engine under-uses it. It is deliberately not acted on, for reasons Section 6.1 gives in full.

Two corrections for the same evidence should not be expected to add. The defusal term the project already ships [12] and the calibration fix this paper derives are, at bottom, one reordering reached two ways — an argument from mechanism, which the measurement then agrees with: stacked, they measure lower than either alone. The ordering is suggestive rather than resolved at this bank size, for the reasons Section 9.2 sets out. The design point stands regardless of how the ordering settles, and Section 9.4 states it once, in general.

2 The rules the inference rests on

Literature is played by six players in two teams of three (seats $0, 2, 4$ versus $1, 3, 5$) [4]. Under `us54` [15] the deck is 54 cards partitioned into nine *sets* (half-suits) of six; nine cards are dealt to each player; the game ends when a team has been awarded five sets. The engine is pure, deterministic TypeScript, and one seed reproduces a byte-identical game [17, 5].

The inference in this paper uses a handful of rows of the `us54` decision table [15], cited by number throughout.

row	rule
5	**Ask target.** The target of an ask is always an opponent; a teammate may never be asked.
6	**Ask licence.** An asker must hold at least one card of the asked set.
7	**Own card.** You may not ask for a card you hold.
9	**Hit.** The card transfers; the asker keeps the turn and asks again.
10	**Miss.** The turn passes to the player who was asked.
12	**Declare content.** Name the set and the exact holder of all six cards; every assignment must be a seat on the declarer's own team.
17	**Information.** Card counts are public, with a full public log of asks, results, and declares.

Two consequences do most of the work below. First, rows 6 and 7 make every ask a *broadcast*: it publishes permanently that the asker holds at least one card of the set, and that the asker lacks the named card. Second, rows 9 and 10 make every ask a *bet with a tempo stake*: a hit takes a card and keeps the turn, a miss ends the turn and hands it to a named opponent. The companion document treats the broadcast and the concession [12]; this paper treats the bet, and what the broadcast lets a listener conclude. The public channel both of them read from — what a log of asks, results and declares does and does not expose — is specified in [10].

3 The direct answer: an absence is not an event

3.1 Why the engine cannot make the inference

FishAI never treats silence as evidence. After A asks B for the 3♠ and misses, the engine concludes exactly one negative fact — B *lacks the* 3♠ — and nothing whatever about B's 2♠/4♠/5♠/6♠/7♠. B's later asks into other sets move those beliefs by zero [11].

The reason is structural rather than careful, and the distinction matters for how much confidence the result deserves. The ingestion routine in `knowledge.ts` is a `switch` over events that *are* in the log [17]. There is no scan for "who has not asked into set S", no silence counter, no candidate-mask decay. An absence is not representable in that layer, so it cannot be acted on there. A guarantee of that shape is stronger than a guarantee that rests on a careful predicate, because there is no predicate to get wrong.

3.2 Demonstrated, not asserted

The claim is executed rather than argued: a treatment in which the 3♠ ask happens against a control in which it does not, with seat 1 then asking four more times and never into low spades [11].

Table 1 is the whole answer to the literal question. The named card's candidate set loses seat 1 — which is row 7 and the logged miss, both sound — and the other four rows are bit-identical between the two runs. Four further asks by the silent seat, into other sets, move nothing.

3.3 The residual runs the other way

The interesting part is what the engine believes in that position, measured against ground truth. In the owner's exact case — I missed on this seat, and it has never asked back into the set — the engine believes 14.0% and the truth is 2.0% (N = 19,003) [11].

Table 1: Candidate sets and per-seat probabilities after the treatment ask, against the control that omits it [11]. Only the named card moves.

card	candidates (treatment)	candidates (control)	p(seat 1) treat.	p(seat 1) ctrl.
3♠	$[2, 3, 4, 5]$	$[1, 2, 3, 4, 5]$	0.0000	0.2000
5♠	$[1, 2, 3, 4, 5]$	$[1, 2, 3, 4, 5]$	**0.2000**	**0.2000**
6♠	$[1, 2, 3, 4, 5]$	$[1, 2, 3, 4, 5]$	**0.2000**	**0.2000**
7♠	$[1, 2, 3, 4, 5]$	$[1, 2, 3, 4, 5]$	**0.2000**	**0.2000**

It is *too generous* to the silent seat, by twelve points, not too harsh. It never writes that seat off; if anything it should write it off slightly more than it does. So the owner's principle is right, the engine already honours it, and it honours it with room to spare. Section 6 takes that twelve-point gap seriously as a quantity, and Section 6.1 explains why the project declines to spend it.

4 Every way a candidate can be eliminated, and whether it is sound

An engine that does not infer from absence can still infer unsoundly from presence. The audit below walks every path by which a candidate seat is removed from a card's candidate set, and asks of each what licenses it [11, 17].

Table 2: The elimination-path audit [11]. Not one trigger is keyed on an absence.

trigger	justification	verdict
a miss — the target is cleared for the **named card**	the logged event	**sound**
a miss — the asker is cleared for the named card	row 7: you may not ask for a card you hold (correctly disabled when `askOwnCardAllowed`)	**sound**
historical count exhaustion	pigeonhole on log-replayed counts; disabled when the log is windowed	**sound**
current count exhaustion	pigeonhole on the public counts of row 17	**sound**
own-hand injection	the viewer's own hand is ground truth	**sound**
singleton collapse, count forcing, constraint forcing, claim reveal	positive fixes, downstream of the above or read from a claim's revealed holders (row 12)	**sound**

Three notes on Table 2.

The bounded arm inherits soundness rather than risking it. The v1.5 bit-budget fact pool [7] carries `lacks-card` and `no-basis` facts, which are compressions of already-certified eliminations; and forgetting can only *widen* a candidate set, never narrow one. A bounded seat is therefore less certain than a full-memory seat and never differently certain.

The refined probability's only failure mode is over-confidence. `refinedHitProbability` short-circuits at zero and is monotone non-decreasing, so it cannot manufacture a false negative — it cannot be the mechanism that writes a seat off. Section 5 then measures whether it is over-confident in practice, and finds it is not.

Two places are sound-but-fragile, and neither is reachable at the shipped hard tier. A first-write-wins rule could persist a wrong certainty on *inconsistent* input, but is reachable only under the easy tier's six-event window, which also runs without constraints; and `clearCand` refuses to empty a candidate set, which is the anti-unsoundness guard itself, erring toward a superset. **No bug was found and no code change is warranted** by this audit [11].

4.1 The one defect the audit did surface

The audit is about elimination, and elimination is sound. The defect it surfaced is about *retention*, and it is the cause of the largest gap in this paper.

`knowledge.ts` records an ask as a deal-time set constraint — *this seat holds at least one card of this set* — and **drops that constraint the moment it is satisfied or exhausted** [17, 12]. Read plainly: the constraint is discarded exactly when the seat has been *shown* to hold a card of the set, which is precisely when the fact is most informative. `refinedHitProbability` iterates those constraints, so it goes blind where the evidence is strongest.

This is not a soundness bug. Nothing false is concluded; something true is forgotten. It is nonetheless the origin of the 8.4-point under-pricing of Section 7.1, and the threat model already routes around it: the header of `threat.ts` names the same staleness and reads the licence off the public log instead, which roughly doubles the share of harvest threats a seat can see (6.0% → 11.7%) and takes detection of the canonical five-and-one position from 7.1% to 35.0% [12, 17]. The fix had never reached the probability.

5 Is the engine over-inferring anywhere? Calibration says no

The audit establishes that each elimination path is licensed. It does not establish that the resulting probability is honest, and the two are different claims: an engine can reason from sound facts to over-confident numbers. So every legal ask at every ask decision was scored with the number `pickAsk` actually decides on, against ground truth read from the hands — 300 `us54` games, $N = 1{,}395{,}876$ scored asks [11].

Table 3: Ask calibration over 1,395,876 scored asks, 300 `us54` games [11]. "Believed p" is the number `pickAsk` decides on; "observed" is the realised hit rate of those asks against the hands.

believed p	N	mean p	observed	observed $-$ p
0.0–0.1	126,303	0.007	0.0068	+0.0002
0.1–0.2	454,977	0.171	0.1681	−0.0028
0.2–0.3	717,054	0.221	0.2023	−0.0184
0.3–0.4	37,765	0.337	0.2912	−0.0461
0.4–0.5	24,124	0.440	0.4746	+0.0349
0.5–0.6	10,083	0.533	0.5205	−0.0130
0.6–0.7	8,181	0.625	0.6499	+0.0249
0.7–0.8	631	0.734	0.7242	−0.0095
0.8–0.9	78	0.810	0.7308	−0.0790
0.9–1.0	**16,680**	**1.000**	**1.0000**	**+0.0000**
all	1,395,876	0.2062	0.1953	−0.0109

Two readings of Table 3, in order of load.

The certainty tier is exact. 16,680 asks called certain hit 16,680 times, no exceptions. That is the property the whole policy is allowed to rest on — row 9 keeps the turn on a hit, so a certain hit is riskless material *and* free tempo — and it holds at this scale without a single counterexample. It is also the strongest available answer to the audit's question, because a false elimination that mattered would have to show up here as a certain ask that missed.

The overall bias is optimistic by about a point (−0.0109: the engine's mean belief exceeds the realised rate). That sign is the signature of an engine that is *not* over-inferring in the feared direction. An engine that wrongly concluded "they must have none" would concentrate its remaining probability on the seats it had not written off, and would therefore show hit rates *above* its own claims in the low buckets. Table 3 shows the opposite in the three low buckets that carry the mass — 0.1–0.2, 0.2–0.3 and 0.3–0.4, 1.21 million asks between them, all negative. The one low bucket that runs the other way is 0.0–0.1, at +0.0002: two ten-thousandths, on asks the engine already calls near-impossible, which is not the concentration the feared error would produce. Among the buckets with mass, the two most negative are 0.3–0.4 (−0.0461) and 0.2–0.3 (−0.0184), which is exactly where the under-priced licence of Section 7.1 lives; the single most negative row in the table is 0.8–0.9 at −0.0790, but it rests on 78 asks and carries no weight.

The estimator here is a reliability decomposition of the familiar kind [1, 2]: the engine's stated probability is the forecast, the realised hit is the outcome, and the bucketed table is the reliability curve. Nothing is recalibrated on the strength of it in this paper; Section 9 explains why the obvious recalibration is declined.

6 What silence is actually worth

Section 3 shows the engine does not use silence. This section asks what it would be worth if it did. Two confounds are removed first: contained sets are excluded, because the contained-book turn-pass deliberately manufactures repeat misses at seats that provably hold nothing, and reuses the same card while doing it [13, 8]. Leaving them in would fabricate the effect the measurement is looking for.

Table 4: Ask-level calibration by silence stratum, contained sets excluded [11]. The third row is the owner's exact position.

stratum	N	believed	observed	error
no prior miss, target **never asked** into the set	1,176,652	0.2007	0.1927	**−0.0081**
no prior miss, target **has** asked into the set	88,419	0.3473	0.4181	**+0.0708**
1 miss by me, never asked back (the owner's case)	19,003	0.1399	0.0199	**−0.1200**
1 miss by me, target has asked into the set since	21,726	0.3712	0.3792	+0.0080

Table 5: The set-level question directly: does the seat hold *any* card of the set? [11]

position	$P(\text{holds} \geq 1 \text{ of the set})$	N
no miss, silent	**0.663**	263,016
miss, then asked back	**0.761**	6,180
miss, then silent	**0.063**	5,968
two or more misses, then silent	**0.004**	464

Three conclusions follow, and they are not the same conclusion.

1. Silence on its own is genuinely near-uninformative. Row 1 of Table 4 is 0.8 points of bias over 1.18 million asks — the largest stratum in the study, and the one that corresponds to "they simply have not asked into this set". The owner's principle is validated at scale: you cannot read anything into "they didn't ask back", taken alone.

2. Silence *after a miss* is a different object. A seat you missed on that then stays quiet holds a card of that set 6.3% of the time (Table 5, row 3), against 66.3% for a silent seat you never missed on. Two or more misses then silence takes it to 0.4%, on 464 observations. The compound event is strongly informative even though its parts are not, and the ask-level view agrees: the engine believes 0.1399 where the truth is 0.0199. *The engine under-uses this signal rather than over-using it.*

3. The engine is not making the mistake the owner was guarding against, in either direction that matters. Its one large error in this table is generosity. Note also row 2 of Table 4, which is the mirror image and is the subject of Section 7.1: where a target *has* published a licence, the engine under-believes by 7.1 points.

6.1 Why the compound signal is not acted on anyway

Twelve points of ask-level bias and sixty points of set-level probability look like an opportunity. The project declines it, and the reason is a measured one rather than a temperamental one.

It is the same trap the companion document records [12]: a plausible small signal read off the public log — refuse to ask the opponent who would punish the conceded turn — cost up to 11.7 points of win rate when acted on, because the row-6 confound made the information point the wrong way. Threat and opportunity are the same fact under row 6, so the seat that is most dangerous to concede to is also the seat most worth taking a card from. Silence is the *weaker* of the two signals, and it is read off the same channel.

If it is ever attempted, the discipline is fixed in advance and is the lab's own [16, 5]: duplicate deals, a byte-for-byte control that reproduces the shipped policy exactly when the mechanism is switched off, and an information-free perturbation of matched magnitude as the null. The last of those is the part that the off-limits work established was necessary: without it, a mechanism that merely *changes* decisions can be mistaken for a mechanism that improves them.

7 Two real gaps, and one pattern that looks like a third

Neither gap is the one the owner feared. Both are its opposite — the engine under-reads *positive* evidence.

7.1 A published licence is under-priced by eight points

When the target has publicly asked into the set — a live row-6 licence — the engine believes 0.2386 and the truth is 0.3221 ($N = 23{,}345$) [11]. It under-prices its best asks by 8.4 points.

The cause is the staleness of Section 4.1: the deal-time constraint that records "this seat holds a card of this set" is dropped the moment it is satisfied or exhausted, and `refinedHitProbability` iterates exactly those constraints.

Conditioning fixes it, and the conditioning is elementary. A live licence says the target holds at least one unresolved card of the set, so under the engine's own per-card estimates q_j for the cards j of set B at target t,

$$\Pr\big(c \text{ at } t \;\big|\; \text{at least one of } B \text{ at } t\big) \;=\; \frac{q_c}{1-\prod_{j\in B}(1-q_j)}. \tag{1}$$

Applied at strength λ — $\lambda = 0$ leaving the shipped probability untouched, $\lambda = 1$ applying Equation (1) in full — the residual bias on the licensed subset runs:

$\lambda = 0.60$ removes the bias almost exactly. That the correct strength is not 1 is itself informative: the licence is a fact about the deal, and by the time it is read the seat may have been partly harvested, so the full conditioning over-corrects. The ladder is monotone in λ and crosses zero once, so the

Table 6: The λ ladder: residual calibration bias on the licensed subset [11].

λ	0	0.25	0.40	0.50	**0.60**	0.75	1.00
bias	-0.0835	-0.0488	-0.0280	-0.0141	$-\mathbf{0.0002}$	$+0.0206$	$+0.0554$

operating point is not a fitted constant so much as a read-off. Section 9 reports what $\lambda = 0.60$ bought in play, which is the part that matters and is not what one would guess.

7.2 The conceded turn is priced with the wrong sign

The second gap belongs to the companion document [12] and is repeated here because it belongs in the ask matrix. `missTarget`, `aimedTarget`, `valueContainedPass` and `signallingAsk` all rank the seat that receives the turn by **hand size**, on the stated grounds that the surrendered turn is worth least at the smallest hand. Over 12,376 observed concessions, hand size correlates $-\mathbf{0.147}$ with the cards that seat then actually takes — negative in every game-phase bucket — against $+\mathbf{0.421}$ for its published reach, the sets it has shown a row-6 basis in [12].

The two gaps have the same root. Both are failures to use the row-6 licence: one on the probability side, one on the concession side. The concession side has a fix that ships (`defuse.ts`); the probability side has the fix derived in Section 7.1 and not shipped, for the reason Section 9 measures.

7.3 Why the same two or three sets get traded — and why it is not a bug

The owner also reported watching games in which "two or three half-suits get ask-traded until it's declared, while the other half-suits never show up". Both halves of that report were tested, and they come apart [11].

The trading is real, and large. After seat B is asked for a card of set H, misses, and takes the turn under row 10, B's next ask goes into H **88.1%** of the time when row 6 lets it ($N = 9{,}091$, 600 games). The availability-matched control — every other legally-askable set at the same decisions — is **21.4%** ($N = 203{,}723$). The provoking set beats the control 5–6× in every bucket, so this is not an artefact of what happens to be askable. Unconditionally the figure is 36.0%, because 59% of provocations land on a set B holds no card of, where row 6 forbids the reply outright.

But it is not a reciprocation rule. It is the certainty rule.

Table 7: What the reply into the provoking set actually is [11].

of B's replies into the provoking set H	share
replies into H that are the argmax on hit probability	**99.1%**
replies into H that are **certain hits**, after a miss	60.2%
… after a hit	94.3%
B would have chosen H *anyway*, before being provoked	**70.1%**
decisions the provocation actually switched into H	**18.1%**

An ask into H publishes that the asker holds a card of H (row 6) and lacks the named one (row 7). Combined with what B already knows, that very often *locates a card exactly* — and `certaintyBonus` makes a certain hit dominate the sort, while row 9 keeps the turn. **The bot is not asking back; it is taking a card it can see, and the ask it was just asked is what made it visible.** It could not be doing anything else: the tradition bans prearranged conventions [3], and nothing in this engine decodes an incoming ask as a signal — only as the two facts rows 6 and 7 make it publish.

Ablations confirm the ordering, and they are decision-level rather than outcome-level — the unit the project holds to be the honest one for a question of this shape [9]. The 88.1% falls to 84.4% with `defuse: 0`, to 72.3% with refined inference off, and to 72.0% with constraints off — still 3.4× the control. The defusal term contributes about four points of it and is not the cause; the `-nodefuse` arms reproduce committed HEAD byte-for-byte on the ask path, which is what licenses that conclusion [11].

The "other sets never show up" half is false at game scale and true locally. Per game the top three sets take 49.2% of asks against a uniform 33.3%, an HHI of 0.1358 against 0.1111 — real but modest — and only 0.55 of 9 sets receive no ask at all. But in any window of 10 consecutive asks only **2.96** distinct sets appear, against **5.98** for the same games' asks randomly permuted, and 75.3% of windows show three or fewer sets against 0.4% permuted. The owner is reporting a local view accurately: the clustering is *temporal*, not a coverage failure. Every set does get its turn; they just take turns in long runs.

Does the clustering cost anything? Not identifiably. The raw correlation between a game's ask concentration and its set outcome is −0.443, but ask *count* correlates +0.929 with sets, and controlling for it collapses the partial correlation to −0.074 (−0.008 on the roster arm). Ask count is plausibly a mediator as well as a confound — games that run long have more asks *and* more sets — so this identifies nothing causal in either direction, and no intervention is warranted on this evidence. The honest statement is that the observed behaviour is explained, not that it is optimal.

8 The holistic ask matrix

The owner asked for the whole reasoning matrix for choosing an ask, not only the silence question. Table 8 is it: twelve factors, each with the rule it rests on, the sign it should carry, and whether the engine prices it today [11].

Summarised: **priced well**, factors 1, 2, 4, 7, 9, 10, 11; **wrong sign**, factor 5, with factor 8 inheriting it; **under-priced**, factor 3; **correctly ignored**, factor 12. The matrix's own headline is that the engine's errors are concentrated in one place — what a published row-6 licence is worth — and that they all run in the direction of under-use.

8.1 The one thing this matrix must not say

The owner pairs "whom to ask" with "keeping in mind the off-limits rule": the idea that some opponents should be off limits because the turn you concede on a miss would be punished. This project's measured position is that the off-limits *prescription* — refuse to ask the dangerous seat — **loses** 4.5 **to** 11.7 **points of win rate**, with the paired set-difference falling monotonically with appetite [12]. Row 6 is the reason: an opponent is dangerous *because* it has published a basis, and a published basis is also the strongest available evidence that an ask into that set will hit. Threat and opportunity are the same fact.

The threat *model* is kept and the prescription inverted — ask them, to strip the card their reach rests on — which is `defuse.ts`. Two measurements of it are on record and both belong here: the prototype term, hooked into a mirror of `pickAsk`, is worth +1.50 and +1.65 sets per duplicate pair on two 600-pair held-out banks, while the *shipped* implementation re-measured on those same two banks at 400 pairs lands lower, at $+1.43 \pm 0.32$ and $+1.58 \pm 0.32$; the gap sits inside the intervals and is what a smaller bank plus a real integration should look like. The gain is replicated in a foreign engine [12, 14]. The only asks for which the original avoidance rule is right are the ones that

Table 8: The ask matrix [11]. "Priced today?" is a statement about the shipped engine, and the two entries that are not "yes" are the two gaps of Section 7.

#	factor	grounding	sign	priced today?
1	**hit probability**	rows 9/10 — a hit takes a card and keeps the turn; a miss ends it	+, dominant	**yes, well.** `wHit`·p is 70 of 100 of the score; overall bias −1.1 pt; the certainty tier is exact
2	**certainty vs. probability**	a certain hit is riskless material *and* free tempo	+, must dominate	**yes.** `certaintyBonus` ≥ 20 required of every style; the contained pass refuses to displace one
3	**target's published licence in the asked set**	row 6 is the only ask licence, so an ask publishes "I hold one of these", permanently	+ on both the probability and the value of taking the card	**partly — the biggest gap.** The *choice* is priced (`defusalBonus`: +1.50/+1.65 sets for the prototype term, +1.43/+1.58 for the shipped implementation). The *probability* is not: §7.1, −8.4 points
4	**what my ask publishes about me**	rows 6 and 7, both permanent	–, small	**yes, as a tie-break only** (`leakEpsilon`). Deliberately narrow: the best ask into a strong set is usually the ask that completes it
5	**who receives the turn on a miss**	row 10	– on the miss branch — but see 6	**wrong sign.** §7.2
6	**threat and opportunity are the same fact**	row 6 makes it structural; threat correlates +0.249 with the best available p, whose mean is 0.471 at high-threat seats against 0.326 elsewhere	**the naive veto is negative**	**yes, correctly.** Avoidance costs up to 11.7 points; `defuse.ts` inverts it
7	**defusal credit**	a hit strips the licence and row 9 keeps the turn	+	**yes**, `defuse: 1` across the roster, replicated in a foreign engine [14]
8	**is the set contained?**	no opponent can ask into a one-team set, and declaring it hands it back	**flips the calculus**: the ask stops being about cards and becomes purely about aiming the turn	**yes, own branch** — but its `gain` is built on factor 5's backwards proxy
9	**progress toward the set**	row 12 needs all six holders named	+, modest	**yes**, `wProgress`
10	**narrowing value of a miss**	row 17 makes every ask public; a miss on a two-candidate card pins it	+, small	**yes**, `wNarrow` — the only term that pays for a miss
11	**clinch state**	the game ends at 5 sets; a wrong declare gifts one	**asymmetric, large late**	**declares yes, asks no.** Nothing in `pickAsk` reads the score. Defensible: an ask cannot lose a set outright
12	**silence**	not a rule — an absence	–, weak alone (§6)	**ignored, and that is correct**

cannot hit, and those are exactly the asks where the engine still ranks the receiving seat by hand size (Section 7.2).

9 Two corrections for the same evidence, and what stacking them costs

Section 7.1 removes a real 8.4-point bias with a principled, knob-free conditioning. The obvious expectation is that a better probability plays better. It does — *but only if the engine is not already being paid for the same evidence.*

9.1 The design, and why the common baseline is the whole point

All arms mirror `pickAsk` as it now stands, defusal term included, with the zero arm verified to reproduce the shipped policy byte-for-byte — the lab's standard control discipline [16, 5]. The measurement is 400 duplicate pairs, one bank, and — the design decision that carries the result — **one common baseline** (`defuse: 0`, $\lambda = 0$), so that the three treatments are directly comparable to each other and not merely each to itself [11].

Table 9: The three-way substitution, 400 duplicate pairs, one bank, one common baseline (`defuse: 0`, $\lambda = 0$) [11]. Intervals are ± 1.96 SE on the paired set-difference. See Section 9.3 on the 400-pair scale.

arm	win rate	paired set-difference
baseline — neither	50.00%	0.0000 ± 0.0000
defusal alone (`defuse: 1`, $\lambda = 0$)	64.00%	$\mathbf{+1.915 \pm 0.315}$
calibration alone (`defuse: 0`, $\lambda = 0.6$)	65.25%	$\mathbf{+2.068 \pm 0.293}$
calibration alone, over-corrected ($\lambda = 1.0$)	61.63%	$+1.620 \pm 0.327$
both together (`defuse: 1`, $\lambda = 0.6$)	60.63%	$\mathbf{+1.565 \pm 0.317}$

9.2 The result

Three things follow from Table 9, in order of how much they change what one would do.

1. The two mechanisms look like substitutes rather than complements, and the ordering is suggestive rather than established. Either alone is worth about two sets per duplicate pair, and the stack measures below either alone. Taken at face value that says adding the principled fix on top of the shipped heuristic would have made the engine worse.

The claim has to be stated at exactly the strength the design supports, which is less than the face value. **Every interval in Table 9 is on an arm against the common baseline; no interval is published on any *difference* between two arms**, and the differences are what the substitution claim is about. Nor can the arm intervals be differenced by eye: the three treatments run on one bank against one baseline, so their errors are positively correlated and a properly paired arm-to-arm interval would be *narrower* than any naive combination of the two published half-widths. That is the design's strength — a common bank and a common baseline make this a paired comparison, far sharper than measuring each treatment on its own bank and subtracting — but the paired interval was not computed, so the gap is unquantified in either direction.

What can be said about magnitude is discouraging at this N. The larger gap, calibration alone against the stack, is 0.503 sets; the smaller, defusal alone against the stack, is 0.350. Section 9.3 puts each arm's own 80%-power minimum detectable effect at 0.42–0.47 sets, so the smaller gap sits below the floor these arms were measured at and the larger one barely clears it — and the empirical floor at $N = 400$ is itself only pinned to 0.36–0.50. The pairing buys back some of that, and the ordering is consistent across both comparisons and reinforced by the $\lambda = 1.0$ row below, which is why the result is worth reporting. But **it is a suggestive ordering at this bank size, not an established one**, and the honest next step is to re-run the same five arms with the paired arm-to-arm differences and their intervals reported, at a bank where 0.35 sets is comfortably resolvable.

What the design *does* establish is that this question is askable at all. Only a common-baseline design can see substitution: measured each against its own baseline — the natural thing to do, and the thing an incremental development process does by default — the calibration fix looks like a flat $+0.04 \pm 0.41$ rather than like something competing with a term already shipped.

2. The mechanism is one reordering reached two ways. Equation (1) multiplies every card of one set at one seat by the same factor, because the denominator depends on the set and the seat and not on the card. It therefore *cannot* reorder asks within a (set, seat) pair; all it can do is promote licensed asks against unlicensed ones. That is exactly what `defuse` does, by a different route — a bonus on the seats whose published reach makes them worth stripping. Two terms that promote the same asks are not additive, and the $\lambda = 1.0$ row is the same shape seen from inside a single mechanism: over-shoot the promotion alone and the gain falls from +2.068 to +1.620, which is what stacking also produces.

3. The calibration error is real and worth fixing eventually — just not here. `refinedHitProbability` is also read by the declare planner, by `valueContainedPass`, and by any future search or exploitability sweep, and in none of those is a defusal bonus quietly compensating for it. It is a bug fix rather than an appetite; it needs no style knob; it would help every consumer of the probability rather than only the ask ranker.

The recommendation is to ship neither change now. `defuse: 1` stays, on the strength of its gauntlet against opponents that also defuse and its foreign-engine replication [12, 14], neither of which the calibration fix has. The calibration fix is recorded with its λ ladder as the better long-term shape. Swapping the two is a deliberate experiment for its own day, not a change to make in passing.

9.3 The scale caveat, and a refuted explanation

Table 9's numbers come from a 400-pair bank, and that is this instrument's thin end. The project has since characterised the instrument: for arms that differ at every decision the per-pair standard deviation is 3.15–3.44 sets, which puts the 80%-power minimum detectable effect at about 0.47 sets at $N = 400$ and about 0.33 at $N = 800$ [12].

That generic figure must not be applied to this cell by eye, and the companion document names the substitution as a trap it fell into itself: the minimum detectable effect depends on *that cell's own* standard deviation, which varies across the repository by up to a half [12]. So the floor here is computed from Table 9's own published half-widths. A half-width of h at $N = 400$ implies a per-pair SD of $h\sqrt{N}/1.96$, giving 3.21 for the defusal arm, 2.99 for the calibration arm, 3.23 for the stack and 3.34 for the over-corrected arm; the corresponding 80%-power minimum detectable effects are 0.45, 0.42, 0.45 and 0.47 sets. Each arm clears its own floor by three to five times, so each arm's effect against the common baseline is resolved. What is *not* resolved is the spacing between the arms: see Section 9.2.

Two properties of the instrument at $N = 400$ failed to replicate, and both belong here. The coverage is unsettled — an independent replication found 10.8% false positives against a nominal 5%, pooling to 8.0% with a confidence interval that excludes 5%, and whether that is real under-coverage or noise is recorded as unresolved. So is the empirical floor: a planted effect of 0.393 sets was detected in 84.7% of banks on the first run and 75.0% on the replication, pooling to 81.7%, which leaves the $N = 400$ floor somewhere in 0.36–0.50 and not pinned tighter by that design — the $N = 800$ floor does replicate [12]. Intervals at this scale should be read with both in mind.

More directly: **the +1.92-scale figures on this bank are high.** The companion document re-ran the defusal quantity at 4,000 duplicate pairs through a single code path, and the same edge measures $+1.6225 \pm 0.1016$, replicated independently at $+1.6595 \pm 0.1022$, against Table 9's $+1.915 \pm 0.315$ [12]. The gap is ordinary sampling variation, and the record of how that was established is worth keeping, because the first answer offered was wrong: it was proposed that this paper's mirror harness lacks `pickAsk`'s certain-hit gate and therefore measured a different quantity. That explanation was *tested*, by running the ungated harness, and **refuted** — $+1.6850 \pm 0.3358$, indistinguishable from the gated value [12]. The harness difference is not the explanation; the bank size is.

None of this disturbs the ordering Table 9 exists to establish, because all five arms share one bank, one baseline and one code path, and the substitution result is an ordering among them rather than an absolute level. It does mean the *levels* in that table should be read as this bank's, and the 4,000-pair figures preferred wherever the same quantity has been measured there.

9.4 The general lesson, stated once

A term that corrects a probability and a term that rewards the same evidence are not additive. Measuring the second after the first has shipped will show nothing — the $+0.04 \pm 0.41$ of Section 9.2 is exactly what that looks like — and stacking them can subtract. Any future mechanism that promotes asks at seats with a published row-6 licence is competing with `defuse` for the same effect, and must be measured against a baseline with `defuse` *off* before it can be believed.

The lesson generalises past this engine. Wherever a heuristic has been tuned into a system and a principled correction for the same underlying error is proposed later, the default comparison — new thing versus current system — is the one comparison that cannot detect substitution. The common baseline costs one extra arm and is the only design that answers the question actually being asked.

10 Threats to validity

1. **The soundness audit is a code audit, not a proof.** Table 2 enumerates the elimination paths present in the implementation and licenses each against a rule; it cannot establish that no future path will be added, and it says nothing about paths in other engines. Its strongest empirical backing is the exact certainty tier of Table 3: a false elimination that mattered would have to surface there.

2. **Calibration is measured against this roster's own play.** The 1,395,876 asks of Table 3 come from 300 self-play `us54` games. The distribution of positions is the roster's, so the reliability curve is conditional on the population that generated it. A different opponent population would produce different bucket masses and could produce different biases; nothing here measures that.

3. **The silence strata are observational.** Table 4 conditions on events the engine's own policy produced — which seats it chose to ask, and when. The strata are therefore not randomised, and the 6.3% figure of Table 5 describes seats the engine chose to miss on rather than seats in general. Contained sets are excluded for exactly this reason (Section 6); other selection effects of the same shape are possible and unmeasured, which is one more reason Section 6.1 declines to act on the signal without a duplicate-deal experiment.

4. $\lambda = 0.60$ **is read off one ladder.** Table 6 is a post-hoc sweep on the licensed subset of the same games the bias was measured on. The ladder is monotone and crosses zero once, so the operating point is not fragile in shape, but it has not been validated out of sample and is not offered as a constant to ship.

5. **Table 9 is a** 400**-pair bank.** Section 9.3 states this in full: each arm clears its own detection floor against the common baseline, the levels are high against a 4,000-pair re-measurement of the one quantity that has one, and both the interval coverage and the empirical floor at this N failed to replicate. The paper does not claim the levels, and it claims the ordering only as suggestive: no interval is published on any difference between arms, and the smaller of the two gaps the substitution reading rests on is below these arms' own floor (Section 9.2).

6. **The substitution result is one operating point.** One λ, one defusal appetite, one roster, one bank. That the two mechanisms appear to substitute at $(\lambda, \texttt{defuse}) = (0.6, 1)$ does not establish the shape of the interaction surface elsewhere on it, and no arm here measures a reduced defusal appetite alongside the calibration fix — which is the obvious next experiment and is not run.

7. **Ask clustering is explained, not priced.** Section 7.3 identifies the mechanism behind the 88.1% and shows the concentration is temporal. The partial correlation of -0.074 is explicitly not a causal estimate — ask count is plausibly a mediator as well as a confound — so nothing here says whether the clustering helps, hurts, or neither.

8. **Self-play, one engine, one rule set.** The usual transfer caveats apply in full [5, 6]. The one external check available is the foreign-engine replication of defusal [14], which covers the mechanism of factor 7 and none of the inference results in this paper.

11 Conclusion

The owner's worry was that a strong inference engine would slide from "no evidence they hold one" to "evidence they hold none". The engine does not, and the reason is structural: absence is not an event, and the layer that would have to act on one is a switch over events that happened. The measured residual runs the other way — 14.0% believed against 2.0% true in exactly the position the worry describes — and the calibration table says the same thing over 1,395,876 asks: the engine is optimistic by about a point, which is the wrong sign for the feared error and the right sign for an engine that under-uses what it knows. The certainty tier, the one place where the policy is allowed to rest its whole weight on the inference, is exact over 16,680 asks.

What the study found instead is that the engine's real errors are all one error seen from different angles: it under-uses the row-6 licence. It under-prices the probability of a hit at a seat that has published a basis by 8.4 points, because the constraint recording that basis is dropped the moment it is confirmed. It ranks the seat receiving a conceded turn by hand size, a proxy that correlates -0.147 with what that seat then takes against $+0.421$ for the licence. And the behaviour that looked most like a bug from the outside — two or three sets traded back and forth — turns out to be the engine using the licence *correctly*: 99.1% of those replies are the hit-probability argmax, and the bot is not reciprocating but collecting certainties that the incoming ask made visible.

The result that changed what the project would do is the last one. A principled correction for the under-priced licence is worth $+2.068 \pm 0.293$ sets per duplicate pair on a common baseline; the shipped heuristic that rewards the same evidence is worth $+1.915 \pm 0.315$; both together are worth $+1.565 \pm 0.317$. Two fixes for one error measure lower stacked than singly — an ordering that is suggestive at this bank and not established, because no interval was published on any difference between the arms and the smaller gap is below their own detection floor. The design claim is the firm one: the only comparison that can *see* substitution is one that measures both mechanisms against a baseline carrying neither. The recommendation is to ship neither change now, and to record the calibration fix — with its ladder, its bank size, and the refuted explanation of its scale — as the better long-term shape for whoever swaps them deliberately.

Reproducibility. The engine is deterministic; every number in this paper is a field of a committed probe result in [11] or [12]. The calibration of Section 5 is `scripts/probe-inference.mjs`; the strata of Section 6 are `scripts/probe-inference-strata.mjs`; the λ ladder of Section 7.1 is `scripts/probe-licence.mjs`; the clustering of Section 7.3 is `scripts/probe-concentration.mjs` with its window and outcome companions; and Table 9 is `scripts/probe-licence3.mjs`, whose `0 lic3 0 0` invocation is the byte-exact control that must print 0.0000 ± 0.0000 [11, 17].

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